Cost Effective Machine Vision System for Real-Time Defect Detection and Energy Efficiency in Learning Factories

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Abstract— Machine vision (MV) is a useful tool for industrial settings that aim for real-time data processing. Any industry may use this to monitor workpieces for geometric faults and material abnormalities in real time. Defect detection, workpiece sorting according to physical qualities, and process abnormality analysis are all made possible by using the convolutional neural network (CNN) method on real-time information from a low-cost, basic Raspberry Pi with a camera and the open-source ML platform TensorFlow. Using the planned cyber-physical production method, the cyber world may be connected to the physical units of a Learning factory (LF) to construct a data collection system based on MV. LFs are now often used in various sectors to train workers to handle the new technologies and problems that have emerged as a consequence of technological advancement. This study shows how to use an affordable MV device in a learning manufacturing setting to get data in real-time and save energy. It was discovered that the suggested inexpensive machine vision system can identify surface flaws, colours, and geometric imperfections. The straightforward and inexpensive MV system has improved energy efficiency and decreased overall carbon footprint by 19.36% and 798.22%, respectively, depending on the MV method along the flow. Incorporating action-based learning methodologies into the classroom also improves the quality of instruction. This may reduce the need for humans and operators while assuring less rework, enhanced control, unbiased decisions, and complete quality assurance.

Keywords: Convolution Neural Network, Machine Vision, Machine Learning, Learning Factory.

I. INTRODUCTION

Cyber-physical systems are the product of the next generation of computers made possible by technological advancements in the last decade (CPS). Utilizing robust autonomous microcontrollers that are interconnected, these systems merge digital and physical components into a unified framework [1]. Cyber-physical manufacturing System is the name given to CPS when used in a manufacturing setting. By eliminating maintenance and breakdown difficulties, as well as production bottlenecks, a well-integrated CPPS architecture allows for efficient manufacturing and decreases production lead time [2]. This means that current manufacturing processes will have to undergo a radical technical transformation to accommodate new technology. Although learning factories are now a secure, affordable, and competitive means for engineering students or newly hired workers to gain transversal abilities, this advancement in technology has created a gap in technical abilities between academia and industry. The factories of tomorrow will not be possible without interdisciplinary education and a focus on practical application [3-8]. As a result, a new concept known as the learning factory (LF) has emerged; it is a model of an idealized value chain industrial segment that houses both formal and informal training. LF's ability to replicate a real-world production-based environment for the transdisciplinary development of key competencies has made it an industry standard and a favourite among academic institutions. LF has grown into a crucial innovation tool that allows researchers to optimize the Cyber-Physical Producing System (CPPS) in real time and integrate new technologies to make the production environment more efficient. Prototypes may be created, refined, and evaluated in a large-scale industrial to the infrastructure of the research factory as it pertains to the industry [9-12]. The capacity to learn and system events using patterns, observations, and predictive examination is a defining feature of CPPS in general. Machine Vision (MV) is one example of a technology that may improve data flow and aid decision-making. To identify, categorize, and detect objects, it uses neural networks and machine learning techniques. To recognize and categorize data using pre-trained image models, MV systems use imaging technologies like cameras. A practical instructional dataset for object-oriented detecting in industries is made available via the CPPS data inflow. The CPPS MV system may enhance the system's dependability and resilience. Affordable camera systems that are simple to incorporate into a manufacturing setting have also emerged as a result of recent advances in consumer electronics [13-16]. Among the many uses for MV systems in industry are automation, control of quality, safety features, and monitoring of conditions in real time. Providing a clearly defined outline with regards to the CPPS, a structure for execution, and objectives for competence development becomes required

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because MV systems and CPPS systems are now developing production paradigms. Modern production processes need a two-way flow of information between new ideas and existing expertise. This kind of information sharing facilitates the practical method of skill development, which aids in the creation of new solutions for the expansion of industrial sectors [17-20]. The primary goal of this article is to lay down the groundwork for a learning factory to adopt a reliable and inexpensive MV system solution. Two main objectives are to improve interdisciplinary learning so that students and researchers may acquire and upgrade CPPS information and to evaluate the novel potential in terms of energy and resource efficiency [21-25]. Learning factories use machine vision to accomplish these aims.

II. METHODOLOGY:

The system notifies the PLC by pulse signal transmission if it determines that the likelihood of a workpiece fracture or flaw surpasses a quarter percentage Workpiece-related data is sent via Node-RED to the host system that is linked to the learning factory. Above Node-RED, in a web display window, you can see all the collected data. collected and saved information on each flaw for further examination. The Raspberry Pi's eight-megapixel camera can capture images in 720p at 30 frames per second. A 15-centimeter ribbon cable that joins at the CSI port is used to connect the camera to the Pi..

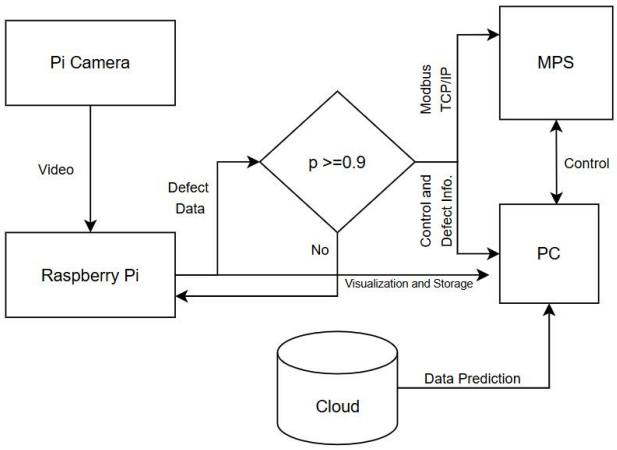

Fig. 1. Flowchart of the machine vision process

The Raspberry Pi receives a live video feed and processes it using a Tensor-Flow Lite model trained on a custom SSD Mobile-net. Tensor-Flow is an open-source, end-to-end machine learning platform that supports a variety of tools and has many resources, such as libraries and community forums. In real-world applications, this model's rapid detection rate and high accuracy rate were the deciding factors. To improve the accuracy of the Tensor-Flow model, it is trained using pictures of both non-faulty and defective workpieces in evaluation and testing scenarios. In the model, there are both faulty and non-defective labels. With nearly eighty per cent of the photos going into training and 20 per cent into testing, an aggregate of pictures has gone into the Tensor-Flow model. It is possible to see the machine vision process flow diagram in Figure 1.

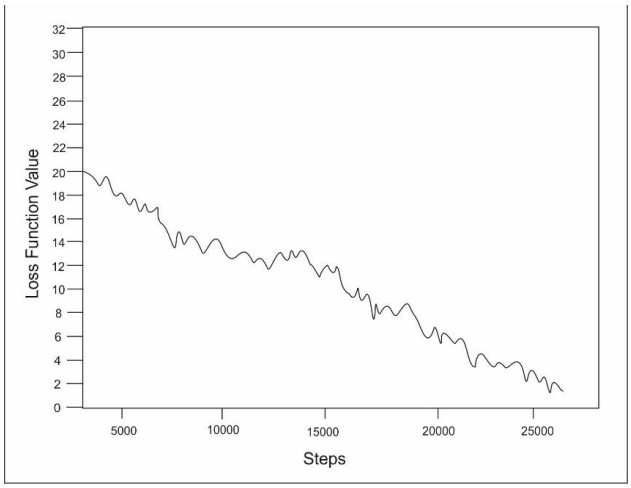

Fig. 2. Data used to train a losing function of value in machine vision

Ensuring that the loss function value stays below 2 requires training the model for around 16 hours. It gets less effective to identify objects when the loss in functional value is much more than 2. Consistently obtaining a loss below two required some steps, as illustrated in Fig. 2. Data from detecting the two types of components in the learning factory's real environment and establishing the model's accuracy is saved in a database. When the chance of finding a flaw is greater than less than half, the Raspberry Pi utilizes a CODESYS add-on to send GPIO signalling pulses to the MPS. Following its connection to the MPS system's control circuit, this signal causes a logic shift that permits label-based workpiece sorting. Along with the constant transmission of this data to the machine's system over LAN, a Node-RED visualization program also continually updates and monitors this data. Using this method, one may track the amount, fault rates, and component geometry in real time. For future data analysis and prediction, the information is saved. This architecture, which takes a human-centred approach, describes the four basic levels of a CPPS and how they interact with each other. The efficient storing and retrieval of data are supported through an internal SQL-based database on the PC, as illustrated in Figure 3.

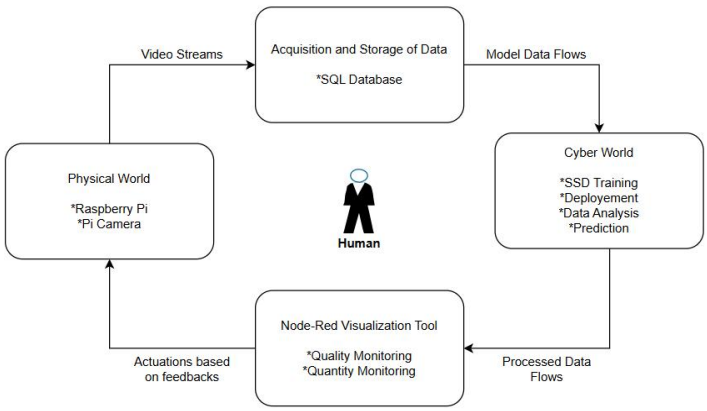

Fig. 3. An Adapted CPPS Framework for the Implementation of Machine Vision Systems

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III. Results

Research and forecasting have shown the current MV system-based CPPS can identify surface quality (the existence of other materials on the surface), geometric flaws (thin and thick fractures), and colour flaws (the existence of other colours). Energy efficiency is enhanced by finding problems early on in the system. The trained model achieves a detection rate of almost 98.5% for a specified class of workpieces. Findings exceeding 85.5% accuracy should be recognized as making it mistake-proof, according to the interpretation of the findings. Therefore, only findings with an accurateness level of 85.5% or above are involved in the database. To train the learned model more often and enhance its accuracy, any frame pictures that fall below the set proportion are added to a second database. The picture shows how the custom-trained model can recognize many objects at once. The implementation of this system also includes a set of educational components. Based on its relevance to the overall system and its absolute need, each learning result is assigned a significance score. In this article, higher-scoring outcomes are prioritized throughout development and execution. With their respective significance scores, the learning outcomes are listed in Table 1.

Table 1. Results of instruction together with their relative significance

S.no	Outcomes from Learning	Score (<= 5)
1	Systems Gathering data	4.5
2	Visualised System of Data	3.5
3	Machine Human Interfaces	4.5
4	PLC Systems	3.5

A. *The Effect of The Visual System on The Parameters of Life Cycle Engineering*

An MV system was shown to be useful in detecting energy and resource losses caused by the scrapping and reworking of faulty workpieces in the learning factory experiment. Instead of wasting energy on rework or moving the workpiece from the distribution station to the sorting station, the system reroutes it if a flaw is formed. This prevents the creation of subsequent defective workpieces. The (BECKHOFF) measuring box and (Twin CAT) software are used to measure energy usage. Power usage in idle and working modes for each MPS are shown in Table 2. Every half a second, the energy is measured. The five stations' energy usage against time plots is shown in Figures 4 and 5. It has been noted that power consumption stays around ten watts while the system is not in use. Power consumption peaks at each station and follows a cyclic pattern when the whole system is operational or in operating condition, including the slider, pneumatic lid pick, and placing arm.

Table 2. Power usage while not in use and when operating in each MPS mode with a 20-second cycle time.

S.no	Stations	Energy Utilisation per cycle (W)
1	Distribution to Stations	286
2	Storage of Stations	704
3	Place and Pick	389 + 321
4	Separation of Data	367
5	Sorting Data	321

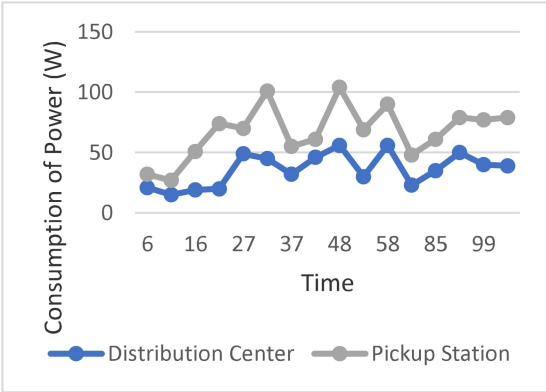

Fig 4. Diagram showing the relationship between distribution and collection station energy consumption and time

Fig 5. The sorting, storage, and separation station's energy consumption shown against time

B. *Analysing power usage with and without an MV system.*

As mentioned earlier, energy losses due to a workpiece's travel through the system might be caused by a faulty one. There will be significant energy waste if the system continues to produce flawed workpieces since this loss will be amplified. The learning factory's third station, the pick-and-place station, has an MV system installed. Without a prompt identification and correction system, all of the power used to transfer defective workpieces to the sorting station would be wasted. This power waste would only occur during the first detection of defective workpieces on the separation station. To illustrate this reduction in power usage, let's pretend that the initial component is flawed and that all subsequent workpieces are also flawed. The MV system can reroute the workpiece any time a flaw is detected, allowing for continuous defect-free production and a reduction in energy consumption as the workpiece flows up to the sorting station (the end of the system). Every step of the manufacturing and distribution processes may benefit from this MV system's fault detection and primary warning capabilities. The energy savings achieved via the adoption of the MV method were calculated using a test scenario involving four workpieces passing through the system. To find the savings that result from two distinct implementation places in the system, it has also been tried to move the MV system. Both the pick-and-place and distribution MPS have their energy usage compared with and without the application of MV in Fig. 6. By immediately identifying the

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defective workpiece in real time, the MV system eliminates this energy waste.

Accounting for an experiment group of four faulty workpieces that are now in the system. To determine the potential energy savings from its installation, two scenarios were considered, one depending on the location of the MV systems and the other on its orientation. Assuming the first scenario, the learning factory's third MPS—the pick-and-place MPS—is where the MV system is integrated. If a flaw is detected, the workpiece is sorted out of the main conveyor at the separating station. Due to the early identification and correction mechanism, this eliminates the energy required to transmit the flawed workpiece through the learning factory. By using the separating conveyor to separate the faulty workpiece, the MV system optimizes power usage downstream. In the second scenario, by looking at the distribution MPS, the first MPS to deploy the MV system. Here, the workpiece is checked for defects before it begins to travel through the system. To achieve even greater energy savings, the conveyor reverses its direction of travel once a fault is detected, diverting the faulty work off the main conveyor.

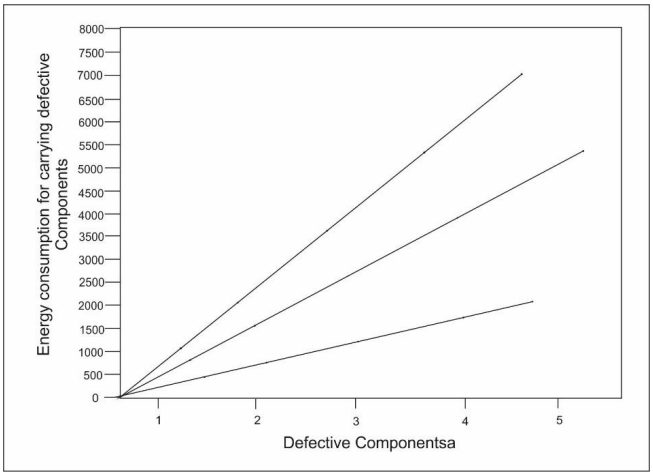

Fig 6. Energy use in the two test situations compared to the basic system without MV

While the basic system (without the MV system) consumes 1721 W when considering a single workpiece, case 2 with the MV system results in a total energy consumption of 1405 W for a sole component in addition to 5485 W for quadruple components (Fig. 6). Saving 317 W, on just one WP is the result of using the MV system in the scenario. The energy savings achieved by implementing the MV system at the distribution MPS are shown in Fig. 6 for the first case. For one WP, the amount of energy saved is determined to be 1347 W. Energy savings as a percentage is:

$$Energy\ Saving = \frac{(Difference\ in\ consumption\ and\ with\ MV\ installation)}{(Energy\ Consumption\ without\ MV\ installation)} \times$$

Following the installation of an MV method at the pick-and-place MPS, the following is the formula for energy savings:

$$Energy\ saving\ \% = \frac{(1721\ -\ 1405)}{1721} \times 100$$

$$= 18.361$$

In a similar vein, the following is the formula for calculating the energy savings upon installation at distribution MPS:

$$Energy\ saving\ \% = \frac{(1721 - 367)}{1721} \times 100$$

$$= 78.675$$

The findings show that a bigger amount of energy may be saved by installing an MV system on the first station. This is because the system can discover defects early on. The energy mix's carbon impact is calculated to be about $671g\ CO_2\ per\ kWh$. For pick-and-place MPS, the estimated carbon footprint reduction from MV system deployment is $0.1g\ CO_2$, while for distribution of MPS, is $0.3g\ CO_2$.

C. *Evaluation of efficiency and cost*

A Raspberry Pi-based MV system may be bought for under ₹10,000 (US $135) a fraction of the price of competing alternatives. The suggested MV system is much more cost-effective than industrial camera modules, which makes them impractical for use in all kinds of inspections in MSMEs. When compared to hiring a single person to do inspections and relying on that person alone, the suggested MV system is much more cost-effective because of its low energy usage and high reliability.

IV. Conclusion

Using a low-cost MV system, the learning factory was able to effectively monitor the workpiece quality for flaws with an accuracy of 85.5% in this article. The article shows how the suggested MV system may help with energy savings. Automated signals were transmitted to activate correction activities in response to three sorts of defects: geometrical, colour, and surface. The use of LF action-based learning methodologies has also improved the quality of instruction and student achievement. Depending on where the MV system is located along the flow, the easy and economical technology has decreased the overall carbon footprint by 78.675% and increased energy efficiency by 18.361%. The whole production may be monitored in real time using the suggested affordable machine vision system. Not only will this allow for less rework, greater control, impartial choices, and 100% quality assurance, but it will also eliminate the requirement for workers/operators. Doing so may assist achieve sustainable manufacturing in any business by improving resource usage and reducing energy consumption. Savings from using the MV system have not been included in this study.

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